

Requirements Specification

Lab Report

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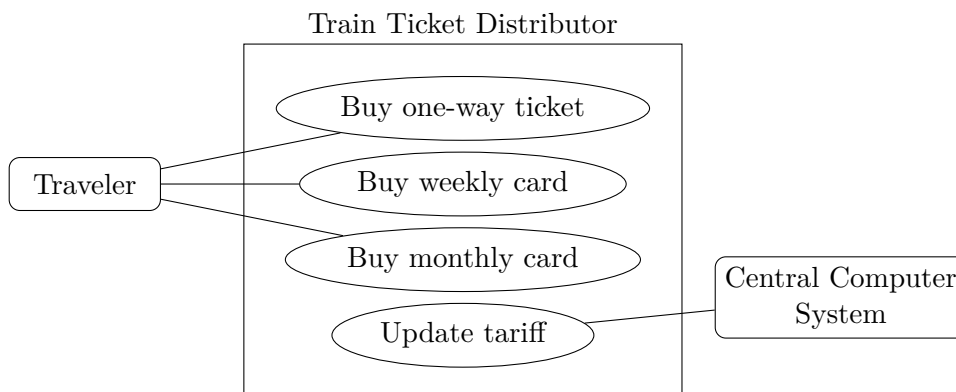
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Warm-up

- 1. Development time.** At 100 lines per day, Firefox (about 10 million lines) would take 100,000 days, or about 274 years. Google services (2,000 million lines) would take 20 million days, or about 54,795 years, assuming 365 working days per year.
- 2. How large systems are built.** They are developed by teams over many years. Teams divide the work into modules, reuse existing components, and coordinate through design, code review, testing, and version control.
- 3. Lines of code and productivity.** The 500-line solution is not automatically more productive than the 50-line solution. Correctness, clarity, maintainability, defects, delivery time, and user value are more useful measures than code volume alone.
- 4. Human contribution with AI-generated code.** Developers still need to understand user needs, define requirements, choose designs, review results for correctness and safety, and take responsibility for the delivered system.

Exercise 2: Train Ticket Distributor

The actors are the Traveler and the Central Computer System. The traveler buys tickets or cards; the central system updates tariffs.

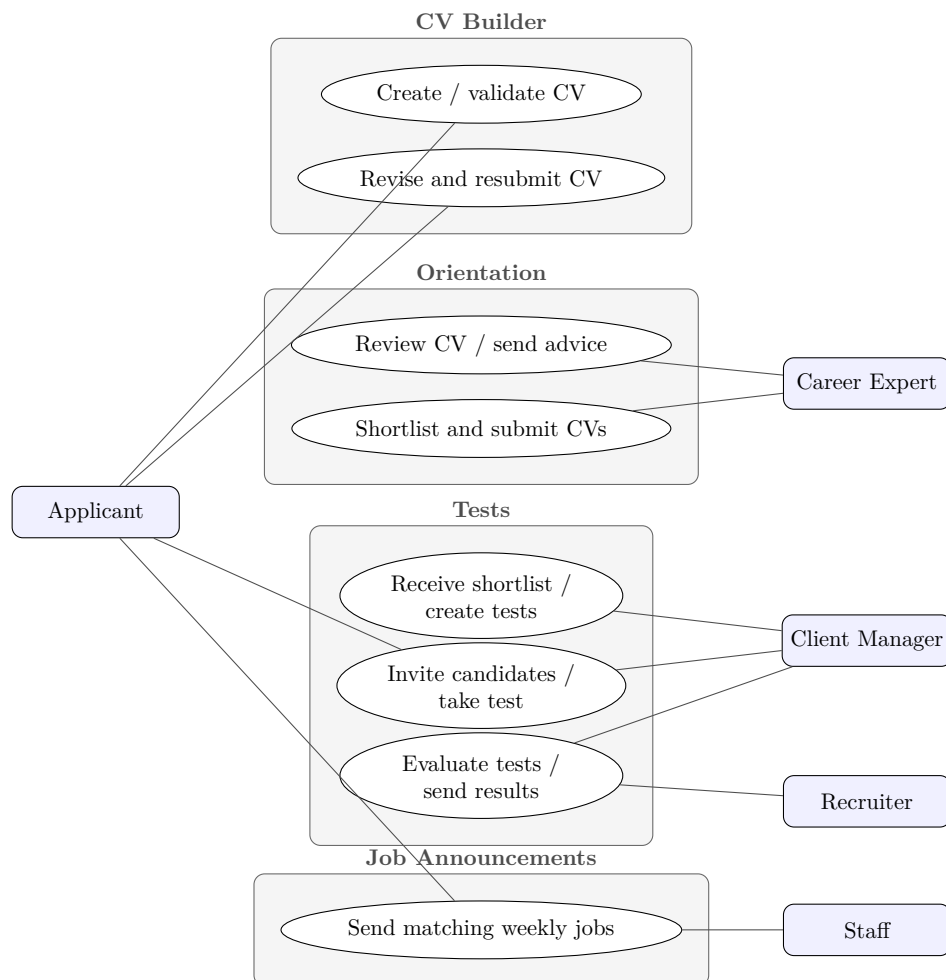


Exercise 3: Recruitment Platform

The actors are Applicant, Career Expert, Client Manager, and Recruiter. The requirements below are functional; estimates are approximate person-days.

ID	Requirement	Priority	Effort
FR1	The applicant can create, validate, and resubmit a CV.	Must	days
FR2	The expert reviews CVs, shortlists suitable applicants, or sends recommendations.	Must	days
FR3	The manager creates tests, invites shortlisted candidates, and the system evaluates completed tests.	Must	days
FR4	The manager sends successful candidates to the recruiter and tells unsuccessful candidates what to improve.	Must	days
FR5	Each week, the system sends users job announcements matching their preferences and profiles.	Should	days

Use-case view (modules). The weekly announcement feature is shown separately because the description does not place it in one of the three named modules.



Exercise 4: Hotel Management System

- Actors.** The hotel client, the receptionist, the cashier, the cleaning staff, the floor manager, and the bank server (an external system).
- Actions outside the system.** The phone call when a client reserves by phone, the client arriving at the hotel, the bank validating the payment, the client eating, the cleaning itself, the floor manager checking the room, and the maintenance repairs.
- Boundaries.** Reservation and Payment, Reception, Billing, and Cleaning.
- Use case diagram.**

